

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr DUMPALA PRASANTH

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

(5 Marks)

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm

converges faster based on the observed results.
(5 Marks)

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. **(5 Marks)**

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. **(5 Marks)**

7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario. **(5 Marks)**

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. **(5 Marks)**

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective. **(5 Marks)**

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning. **(5 Marks)**

Code of Conduct:

*I B.Vani(192472348) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: B.Vani

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

1. Interpretation of Increasing Reward

The reward values increase gradually from **10 to 45** over multiple training episodes. This indicates that the RL agent is learning from its interaction with the environment and improving its decision-making policy.

As training progresses, the agent discovers actions that produce higher rewards and gradually reduces ineffective actions. The increasing reward trend suggests that the policy is becoming more effective and the agent is learning useful state-action relationships.

Therefore, the gradual increase from 10 to 45 indicates **positive learning progress and policy improvement**. If the reward eventually becomes stable around a high value, it would further suggest that the agent is approaching convergence.

Conclusion: The increasing reward trend indicates that the agent is successfully learning and improving its policy over time.

2. Effect of Epsilon in Epsilon-Greedy Strategy

The epsilon values are **0.1, 0.3, and 0.5**, and the cumulative reward is highest at ($\epsilon=0.1$) and lowest at ($\epsilon=0.5$).

In an epsilon-greedy strategy, (ϵ) determines the probability of **exploration**. A higher epsilon causes the agent to select random actions more frequently, while a lower epsilon allows the agent to exploit the best-known actions more often.

At ($\epsilon=0.1$), the agent performs more exploitation and uses its learned knowledge effectively. At ($\epsilon=0.5$), the agent explores much more frequently, which can result in unnecessary actions and lower short-term rewards.

Therefore:

[
 $\boxed{\epsilon=0.1}$
]

provides the best performance among the tested values.

Conclusion: The result shows that exploitation of the learned policy is more beneficial in this particular training scenario than excessive exploration.

3. Q-Learning vs SARSA

Q-Learning and SARSA are compared using average rewards over **100, 200, and 300 episodes**. Q-Learning consistently produces slightly higher rewards than SARSA.

Q-Learning is an **off-policy** algorithm. It updates its Q-value using the maximum estimated value of the next state:

[
 $Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$
]

SARSA is an **on-policy** algorithm and updates using the action actually selected by the current policy.

Since Q-Learning produces higher rewards at all three evaluation points, it

demonstrates better performance in the given experiment. The consistently higher reward trend also indicates that Q-Learning reaches a better-performing policy more quickly in this scenario.

Conclusion: Based on the observed results, **Q-Learning performs better and appears to converge faster than SARSA.**

4. Effect of Discount Factor

The tested discount factors are **0.2, 0.5, and 0.9**, and the average reward increases as the discount factor increases.

The discount factor (γ) controls how much importance the agent gives to future rewards:

- ($\gamma=0.2$): Strong preference for immediate rewards.
- ($\gamma=0.5$): Moderate consideration of future rewards.
- ($\gamma=0.9$): Strong consideration of long-term rewards.

Since the average reward increases with the discount factor, the agent benefits from considering future consequences of its actions.

Therefore:

[
 $\boxed{\gamma=0.9}$
]

is the most suitable among the tested values.

Conclusion: A high discount factor is beneficial in this scenario because successful decisions depend more on long-term rewards than only immediate rewards.

5. Improvement in Success Rate

The success rate increases from **45% at 50 episodes to 92% at 200 episodes.**

This represents an improvement of:

[
 $92-45=47\%$
]

The substantial increase indicates that the agent is learning from its previous experiences and selecting better actions. At 50 episodes, the relatively low success rate suggests that the policy is still developing. By 200 episodes, the 92% success rate indicates that the agent has learned a highly effective policy.

However, a 92% success rate alone does not mathematically prove that the policy is globally optimal. Additional evidence such as reward stability, convergence, and comparison with the best possible performance would be required.

Conclusion: The data strongly indicates significant policy improvement and suggests that the agent is approaching an optimal policy.

6. Effect of Learning Rate in CartPole

The learning rates tested are **0.01, 0.05, and 0.1**, and the agent shows the highest stability at **0.05**.

A learning rate controls the size of each update made to the model:

- **0.01:** Small updates provide stable learning but may make convergence slow.
- **0.05:** Provides a balance between update speed and stability.
- **0.1:** Large updates can make learning faster but may cause instability or overshooting.

Since 0.05 provides the highest stability, it represents the best trade-off between learning speed and convergence stability in this experiment.

[
 $\boxed{\alpha=0.05}$
]

Conclusion: A learning rate of 0.05 is the most appropriate among the tested values because it provides stable and effective learning.

7. Comparison of Exploration Strategies

The three strategies are **ϵ -greedy, softmax, and random exploration**, with ϵ -greedy producing the highest cumulative reward.

ϵ -Greedy: Selects the best-known action most of the time and randomly explores with probability (ϵ).

Softmax: Assigns probabilities to actions according to their estimated values, giving better actions higher probabilities.

Random Exploration: Selects actions randomly without strongly considering learned values.

In this experiment, ϵ -greedy achieves the highest reward because it provides a simple balance between exploration and exploitation. It exploits successful actions while still allowing occasional exploration.

Random exploration is less efficient because it frequently ignores learned knowledge. Softmax can provide smoother exploration, but it may not perform as well as ϵ -greedy for the particular environment.

Conclusion: Based on the observed cumulative reward, **ϵ -greedy is the best-performing strategy** in this scenario.

8. Reduction in Maze Steps

Initially, the RL agent requires **50 steps** to reach the goal. After 300 episodes, it requires only **15 steps**.

The reduction is:

[
 $50-15=35\text{ steps}$
]

The percentage reduction is:

[
 $\frac{50-15}{50}\times 100=70\%$

]

Thus, the agent reduces its path length by **70%**. This indicates that the agent has learned more efficient routes and has improved its decision-making policy.

If the number of steps remains close to 15 for subsequent episodes, it would indicate that the policy has converged. However, based only on the given data, we can conclude that the agent has learned a significantly better policy, while complete optimality requires further evidence.

Conclusion: The reduction from 50 to 15 steps demonstrates substantial learning and suggests convergence toward an efficient policy.

9. Effect of Discount Factor in SARSA

The SARSA agent is tested with discount factors of **0.4, 0.7, and 0.95**, and the highest long-term reward is obtained with **0.95**.

A low discount factor emphasizes immediate rewards, whereas a high discount factor places greater importance on future rewards.

At:

[

$\gamma=0.4$

]

the agent strongly prioritizes immediate outcomes.

At:

[

$\gamma=0.7$

]

the agent considers both immediate and future rewards.

At:

[

$\gamma=0.95$

]

the agent strongly values long-term consequences.

Since 0.95 produces the highest long-term reward, it is the most effective value among the tested options.

Conclusion: ($\gamma=0.95$) is most suitable because the environment benefits from long-term decision-making and future rewards are highly important.

10. Reward Stability in a Stochastic Environment

The RL agent initially experiences fluctuating rewards during the early training episodes but the rewards become stable after approximately **500 episodes**.

The fluctuations in the early stage are expected because the agent is still exploring the environment and has limited knowledge of the best actions. As training continues, the agent collects more experience and improves its policy.

After 500 episodes, the stabilization of rewards indicates that the agent is consistently selecting actions that produce similar outcomes. This suggests that the learned policy has become more reliable and that the agent has adapted to the stochastic environment.

Stable rewards also indicate improved robustness because the agent is less dependent on random exploratory decisions.

Conclusion: The transition from fluctuating rewards to stable rewards demonstrates policy improvement, learning stability, and increased robustness of the RL agent.

Overall Conclusion

The given data demonstrates several important characteristics of Reinforcement Learning. Increasing rewards and success rates indicate policy improvement, while decreasing maze steps demonstrate improved efficiency. The results also show that hyperparameters such as **epsilon, learning rate, and discount factor** significantly influence learning performance. Comparing algorithms and exploration strategies helps identify the approach that performs best for a particular environment. Overall, stable and improving performance over training episodes indicates that the RL agent is learning an increasingly effective policy.